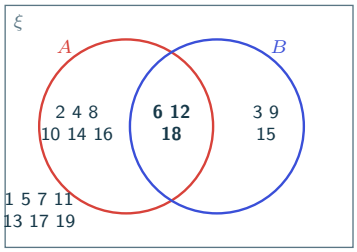
SETS ON ONE PAGE

Set notation: unions, intersections and brackets

From two sets of numbers to the eight regions of three, with the brackets that pick each region out.

1 / THE NOTATION AND A WORKED EXAMPLE

Take $\xi = \{1, \dots, 19\}$, $A =$ even numbers, $B =$ multiples of 3



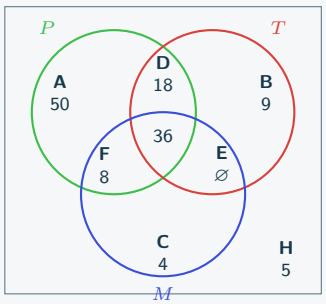
$A \cap B$ in the middle

symbol	read as	meaning
ξ	universal set	everything under discussion
$x \in A,\ x \notin A$	is, is not an element of	$6 \in A,\ 5 \notin A$
$A \subseteq B$	subset	every element of A is in B
$\varnothing$	empty set	no elements
A'	complement	in ξ but not in A
$A \cup B$	union	in A or B or both
$A \cap B$	intersection	in both A and B
$n(A)$ or $\lvert A \rvert$	number of elements	$\lvert A \rvert = 9$
$\{x : \dots\}$	set builder	$A = \{x \in \xi : x \text{ even}\}$

$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$, $B = \{3, 6, 9, 12, 15, 18\}$, $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18\}$ and $A \cap B = \{6, 12, 18\}$, with $\lvert A \cup B \rvert = \lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert = 9 + 6 - 3 = 12$ and $(A \cup B)' = \{1, 5, 7, 11, 13, 17, 19\}$.

2 / THREE SETS / THE EIGHT REGIONS

Let $\xi = \{1, \dots, 100\}$ and ask what belongs in each region



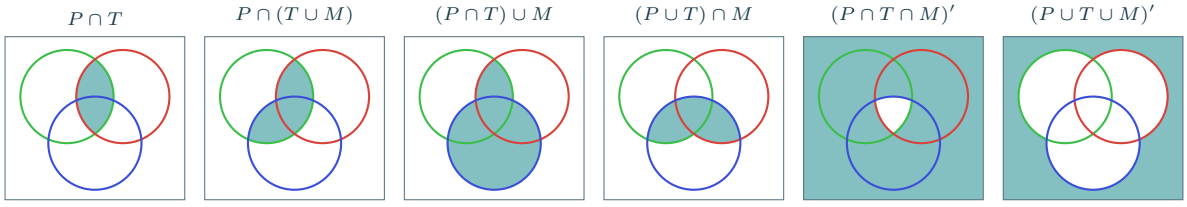
region	expression	example	members
A	$P \cap (T \cup M)'$	$50 = 2 \cdot 5^2$	3
B	$T \cap (P \cup M)'$	$9 = 3^2$	12
C	$M \cap (P \cup T)'$	$4 = 2^2$	1
D	$(P \cap T) \cap M'$	$18 = 2 \cdot 3^2$	13
E	$(T \cap M) \cap P'$	none	0
F	$(P \cap M) \cap T'$	$8 = 2^3$	16
centre	$P \cap T \cap M$	$36 = 4 \times 3^2$	8
H	$(P \cup T \cup M)'$	5	47

Here P is "a product of at least 3 primes" (repeats count, so $8 = 2 \cdot 2 \cdot 2$), T is "has 3 as a factor" and M is "a multiple of 4". **Region E is empty:** a number divisible by 3 and 4 is a multiple of $12 = 2 \cdot 2 \cdot 3$, which already has three prime factors, so $T \cap M \subseteq P$. **Region C is just $\{4\}$:** a multiple of 4 has two 2s, and staying out of P allows no third prime factor.

$T \cap M \subseteq P$, so $(T \cap M) \cap P' = \varnothing$; $\quad P \cap T \cap M = \{12, 24, 36, \dots\}$, the multiples of 12

3 / BRACKETS / THE SAME LETTERS, DIFFERENT REGIONS

The shaded region changes when the brackets move



21 members | 37 members | 38 members | 24 members | 92 members | 47 members

Moving one bracket turns $P \cap (T \cup M)$ into $(P \cap T) \cup M$, which gains the single extra member 4 from region C. The distributive law $P \cap (T \cup M) = (P \cap T) \cup (P \cap M)$ holds, and so do De Morgan's laws, $(P \cup T)' = P' \cap T'$ and $(P \cap T)' = P' \cup T'$.

4 / COUNTING THE UNION

Inclusion and exclusion keeps each region counted once

$$\lvert P \cup T \cup M \rvert = \lvert P \rvert + \lvert T \rvert + \lvert M \rvert - \lvert P \cap T \rvert - \lvert P \cap M \rvert - \lvert T \cap M \rvert + \lvert P \cap T \cap M \rvert$$
$$= 40 + 33 + 25 - 21 - 24 - 8 + 8 = 53, \qquad \lvert \xi \rvert - 53 = 47 = \lvert (P \cup T \cup M)' \rvert$$

Adding the three sets counts the overlaps twice, subtracting the pairs removes the centre three times over, and the triple term puts it back once.

ASIDE / ALWAYS WRITE THE BRACKETS

Unlike $\times$ before $+$, no rule ranks $\cup$ against $\cap$, so $P \cup T \cap M$ is ambiguous and the two bracketings give different regions.

ASIDE / CHECK IN PYTHON

With `P, T, M` built as Python sets over `range(1,101)`, `len(P|T|M)` gives 53 and `T&M-P` gives `set()`.

Convention. Regions A to H follow the slide; the sets are named P, T, M so that the letters do not clash with the regions.

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